How fast does the voice assistant answer?

Sentinel AI incident assistant · Snapdragon 8 Elite dev kit (Sun for arm64 / SM8750P) · build 8909ac4 · 11 October 2026

This report measures the time between a user finishing a spoken answer and the assistant starting to speak its reply, across 23 turns from 6 scripted incident conversations (1 passes), and breaks that time down by stage. Everything ran on the device: speech recognition and incident classification on the Hexagon NPU, the language model and the voice on the CPU.

1778 ms
median, end of speech → reply audible

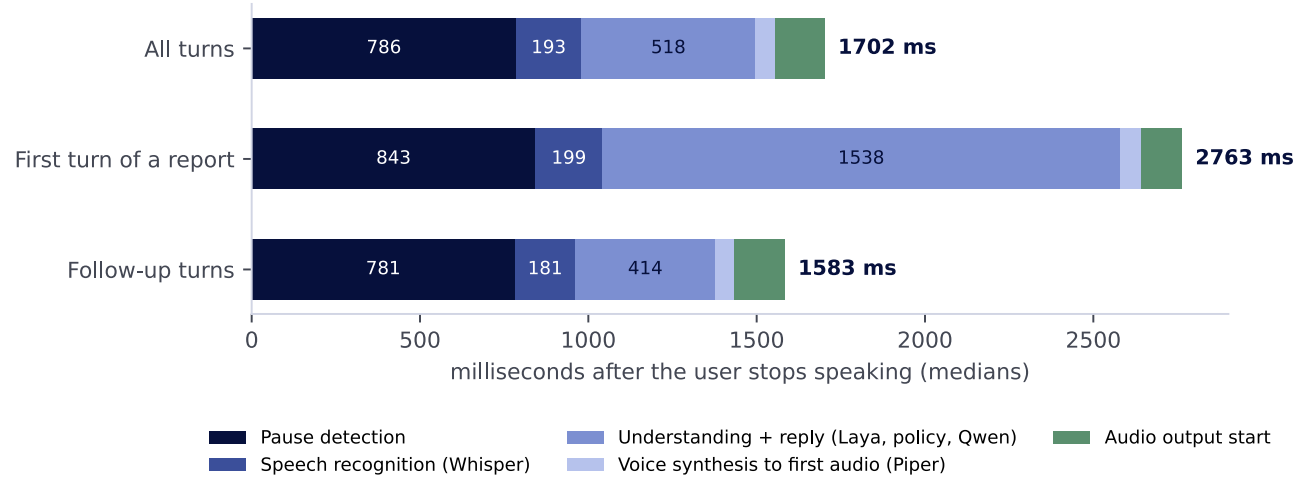
2787 ms
90th percentile (worst 2961 ms)

1002 ms
median after the pause is detected

19/23
turns transcribed word-perfect

In short. A typical spoken turn gets an audible reply in about 1.8 s after the user stops talking (p90 2787 ms). About 786 ms of that is the deliberate pause the assistant waits for to be sure the user has finished; the computing itself takes about 1002 ms. The first turn of each report is slower (median 2732 ms vs 1639 ms for follow-ups) because the language model then reads everything said so far to fill the form in one go. The NPU stages are small: Laya 11 ms and Whisper about 191 ms per turn.

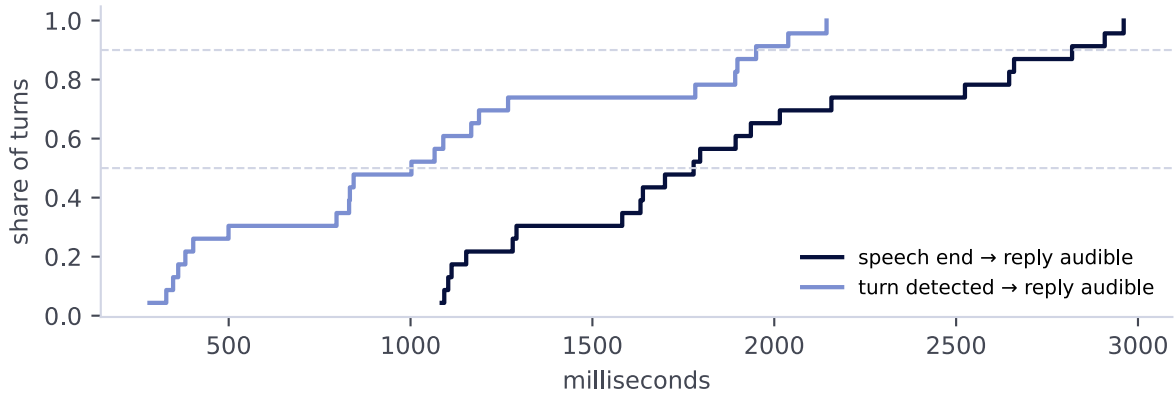
Where the time goes



Median time spent in each stage, from the moment the user stops speaking until the reply's first sound plays. Bar totals are sums of the stage medians, so they differ slightly from the overall median. "Speech recognition" includes waiting for a live-caption transcription that may still be running when the turn ends.

Stage	n	p50 ms	p90 ms	min	max	
Pause detection (end of speech → turn detected)	23	786	871	706	889	0.9 s pause rule, minus the fade of the last word
Speech recognition: Whisper base, encoder on NPU	23	191	223	124	242	10 s window graph for short turns
Incident classifier: fine-tuned Laya W8A16 on NPU	23	11	14	11	14	one inference over all user sentences

Qwen3-0.6B value extraction (CPU)	11	906	1172	312	1207	only when fields are wanted
Qwen3-0.6B question writing (CPU)	14	409	629	325	804	ask/detail turns
Whole assistant step (classifier + policy + Qwen)	23	518	1559	0	1618	
Turn detected → answer text ready	23	712	1739	125	2009	
Piper TTS: text → first audio synthesized	23	57	157	38	190	sentence streaming, 1.2× speed
Text → reply audible (TTS + audio output)	23	207	223	129	293	first frame played (AudioTrack marker)



Distribution over all 23 turns. Read across at 0.5 for the median and 0.9 for the 90th percentile.

What was measured, and how

Voice path, end to end. Each user line was synthesized with the Piper voice and streamed into the app's real voice engine in 20 ms frames at real-time pace, over low background noise (about −66 dBFS), exactly as microphone audio arrives. The same code then runs as in a live conversation: pause detection, Whisper base (encoder on the NPU, decoder on the CPU), the fine-tuned Laya classifier on the NPU, the form policy, Qwen3-0.6B for value extraction and question wording, and the Piper voice for the reply. A turn only continues once the previous reply has finished playing.

Timestamps. *End of speech* is when the last frame of the user's audio entered the engine. *Turn detected* is when the pause detector closed the turn. *Answer text ready* is when the engine published the next question. *Reply audible* is when the first frame of the reply's audio was played by the audio system (an AudioTrack position marker), so it includes synthesis and the audio output buffer, but not the speaker's own acoustic delay (a few ms).

Other metrics. Word error rate of each transcript against the scripted line; app memory (PSS); average CPU cores used during a turn (process CPU time ÷ wall time); battery temperature and Android's thermal headroom (1.0 = throttling).

Accuracy of what was heard

19 of 23 turns were transcribed word-perfect; mean word error rate 2.8%. Building codes are the weak spot: a single letter spoken as a letter ("N building") is easily heard as "M", "an end" or "in". Whisper is now primed with the site's building codes, and a small normalization step maps the common confusions back ("an end building" → "N building", "Q.R.C." → "QRC", a lone "Dunn" → "done"). 5 conversations were filed by voice.

Misheard turns:

- said “It is in N building, on the fourth floor.”, heard “It is N building on the fourth floor.” (WER 0.11)
- said “It is in QRC, on the second floor.”, heard “It is NTORC, on the second floor.” (WER 0.25)
- said “It is in N building, ground floor.”, heard “It is N building, ground floor.” (WER 0.14)
- said “It is in N building, first floor.”, heard “It is N building, first floor.” (WER 0.14)

Only one mishearing mattered. In the four 'N building' turns the error is just an added or dropped 'in'; building and floor were still captured. 'QRC' heard as 'NTORC' left the location unknown, so conversation 2 kept asking for the building and its final 'done' (a very short word) got no reply within the 90 s limit; that report was not filed. The other five conversations were filed by voice.

Resources

Metric	Value	
App memory (PSS)	median 2459 MB, max 2502 MB	all models resident: Laya DLC, Whisper + 2 NPU encoder graphs, Qwen GGUF, Piper
CPU during a turn	median 0.67 cores, max 1.53	out of 8 Oryon cores; Qwen and the Whisper decoder dominate
Battery temperature	24.0–24.0 °C	no sustained heating over the runs
Thermal headroom	0.33–0.40	Android forecast; 1.0 means throttling, so far from it
Model loading (once, at app start)	1413 ms total	tokenizer 349 ms, Laya on NPU 205 ms, Whisper + encoders 385 ms, Qwen 417 ms

What would make it faster

- **Shorter pause rule.** Waiting for 0.9 s of silence is the single biggest fixed cost (786 ms). 0.6 s would save about 300 ms per turn, at the price of cutting off users who pause mid-sentence more often.
- **Fewer or faster language-model calls.** Qwen extraction (p50 906 ms) and question writing (p50 409 ms) run on the CPU. Using the taxonomy's fixed question text instead of a written one saves the second call; the planned fine-tuned Qwen should extract better with shorter output; moving the model to the NPU (Genie) would cut both.
- **Start speaking sooner.** The reply voice already streams sentence by sentence (first audio in about 57 ms); the fixed lead-ins ("Got it, your report is ready…") could be spoken while the language model is still writing the question.

Limits of this measurement

- The user's voice is synthetic and clean. Real voices, accents and a busy room make recognition harder; the timings are representative, the accuracy is optimistic.
- Audio was injected into the engine rather than captured by the microphone, so microphone buffering (about 20–40 ms) is not included.
- One device, one session. Times include whatever else the kit was doing; the 90th percentile shows the spread.

Turns that did not complete

- Conversation 2, turn 4 (“done”): no answer

Appendix: every turn (first pass)

#	User said	Heard	WER	pause	ASR	Laya	extract	question	TTS 1st	total ms
1.1	There is water dripping from the ceiling near the pantry.	There is water dripping from the ceiling near the pantry.	0.00	817	242	11	1136	441	71	**2961**
1.2	It is in N building, on the fourth floor.	It is N building on the fourth floor.	0.11	776	192	11	–	504	169	**1778**
1.3	It is coming from a pipe, and it is spreading.	It is coming from a pipe and it is spreading.	0.00	806	215	11	–	409	91	**1639**
1.4	Yes, there is a socket right below it.	Yes, there is a socket right below it.	0.00	868	221	11	378	–	67	**1700**
1.5	done	Done.	0.00	871	152	11	–	–	45	**1153**
2.1	I can smell smoke near the electrical room.	I can smell smoke near the electrical room.	0.00	869	178	14	1106	446	152	**2819**
2.2	It is in QRC, on the second floor.	It is NTORC, on the second floor.	0.25	706	217	13	312	325	49	**1796**
2.3	No one is hurt, and the alarm is ringing.	No one is hurt and the alarm is ringing.	0.00	848	223	11	318	386	61	**2015**
3.1	The lift is stuck between floors with two people inside.	The lift is stuck between floors with two people inside.	0.00	741	239	14	906	409	51	**2524**
3.2	It is in N building, ground floor.	It is N building, ground floor.	0.14	889	199	11	324	528	158	**2157**
3.3	Yes, two people are trapped.	Yes, two people are trapped.	0.00	786	179	13	399	–	49	**1582**
3.4	done	Done.	0.00	781	134	13	–	–	42	**1281**
4.1	The lights keep flickering in the corridor.	The lights keep flickering in the corridor.	0.00	870	199	11	1207	397	55	**2909**
4.2	QRC building, third floor.	QRC building, third floor.	0.00	870	169	13	–	672	150	**1936**
4.3	Several lights are affected.	Several lights are affected.	0.00	723	160	11	–	–	57	**1104**
4.4	done	Done.	0.00	724	150	11	–	–	38	**1086**
5.1	Someone spilled coffee all over the floor in the cafeteria.	Someone spilled coffee all over the floor in the cafeteria.	0.00	747	194	11	1144	350	52	**2646**
5.2	It is in N building, first floor.	It is N building, first floor.	0.14	789	191	11	–	344	190	**1632**
5.3	done	Done.	0.00	764	124	11	–	–	38	**1093**
6.1	Something is wrong with the water here.	Something is wrong with the water here.	0.00	889	191	11	–	–	99	**1291**
6.2	The toilet is overflowing.	The toilet is overflowing.	0.00	767	160	11	1172	337	51	**2659**
6.3	QRC building, fifth floor.	QRC building, fifth floor.	0.00	706	167	14	–	804	148	**1894**
6.4	done	Done.	0.00	766	134	14	–	–	40	**1113**

Times in ms. "pause" = end of speech → turn detected; "total" = end of speech → reply audible. Raw data and the generator: scripts/latency_report.py, scripts/probe_kit.sh (VOICE=1), app/src/debug/…/VoiceBench.kt.